Communication-Aware Passive Wi-Fi Sensing Under Dynamic Traffic Conditions

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Abstract— Passive Wi-Fi sensing infers human activity and behavior from ambient Channel State Information (CSI) without requiring any dedicated sensing hardware, relying instead on wireless signals already present in a space. While this approach has proven effective under controlled conditions, real environments carry constantly shifting background network traffic, browsing, streaming, uploading, and downloading, that can distort the very signals sensing depends on, in ways that have not been systematically examined. This paper presents a communication aware study of passive Wi-Fi sensing that directly compares it against a controlled active sensing baseline built on the same ESP32 based platform, evaluating both a general activity recognition task and a more subtle intrusion behavior detection task under realistic mixed network traffic. The results show that passive sensing accuracy is closely tied to what is happening on the network at the time of sensing, with bandwidth heavy traffic such as streaming starving the passive receiver of the packets it needs to build a reliable picture of activity, while active sensing remains largely unaffected because it controls its own signal timing. These findings point to network traffic itself as an overlooked variable in passive sensing system design, one that matters as much as sensor placement or model architecture for applications such as forensic monitoring and intrusion detection that must work reliably outside the lab.

Keywords— passive sensing, Wi-Fi sensing, Channel State Information, human activity recognition, intrusion detection, network traffic, ESP32

I. INTRODUCTION

Wi-Fi sensing turns the wireless signals already present in a building into a source of information about the people and activity inside it. Because it reuses existing Wi-Fi infrastructure rather than requiring dedicated cameras or wearables, it has drawn growing interest for applications ranging from fall detection to occupancy monitoring to security [1]. Most of this work falls into one of two paradigms. Active sensing places a transmitter and receiver in a fixed, controlled arrangement so that the timing and rate of the probing signal are known in advance. Passive sensing instead listens to whatever traffic is already flowing across a network, whether it originates from a laptop streaming video or a phone syncing in the background, and extracts Channel State Information (CSI) from those packets without transmitting anything of its own.

This difference matters more than it might first appear. An active sensing link can be tuned so that its packet rate stays stable regardless of what else is happening on the network. A passive receiver has no such guarantee. It only sees a packet when the network itself decides to send one, so its effective sampling rate rises and falls with ordinary daily use of the network, video calls, downloads, idle periods, and everything in between. Despite this being a foundational property of passive sensing, no prior work has systematically measured how different types of background traffic affect the accuracy of the activities it is trying to recognize.

This paper addresses that gap directly. We build an active sensing baseline and a passive, communication aware sensing pipeline on the same ESP32 based platform, in the same physical space, and evaluate both under a set of realistic background traffic conditions generated by a scripted traffic scenario. We apply the passive pipeline to a task with clear practical value, recognizing intrusion relevant behaviors such as a door handle being jiggled or a person dragging an object, purely from ambient Wi-Fi packets and with no dedicated transmitter near the target. Taken together, our contributions are:

- A systematic study of how traffic type (streaming, download, mixed) impacts passive Wi-Fi sensing accuracy, evaluated with a leave one recording out protocol.
- A direct, same platform comparison between active and passive sensing that isolates the effect of losing control over transmission timing.
- A case study applying communication aware passive sensing to a forensic and security relevant task, detecting suspicious intrusion behaviors from ambient traffic alone.

II. RELATED WORK

Lightweight, standalone CSI collection on commodity hardware was made practical by tools such as the ESP32 based platform introduced by Hernandez and Bulut for active repositioning and mobility sensing [2], which forms the basis of the active sensing setup used in this work. Hernandez and Bulut later surveyed the broader challenges of deploying Wi-Fi sensing at the edge, noting that real world signal processing pipelines must contend with noise sources that never appear in curated lab datasets [1], a concern that directly motivates the traffic aware evaluation performed here.

Passive CSI sensing has already been applied to security relevant problems. McDonough et al. proposed Wi-Alert, a passive doorstep sensing system for real time package theft alerts that, like this work, senses suspicious behavior without a dedicated transmitter near the target [3]. Our intrusion behavior detection task follows the same spirit, extending it to door handle jiggling, door opening and closing, and dragging, and evaluating it explicitly against the network traffic conditions present during sensing. Fahad and Bulut have shown separately that WiFi CSI is sensitive enough to detect fine environmental changes such as liquid temperature using physics guided machine learning [5], underscoring how easily CSI reflects factors beyond the target activity itself, and that line of sight versus non line of sight channel conditions can substantially shift sensing performance [4], a channel level parallel to the traffic level effect studied here. Fahad, Touhiduzzaman, and Bulut further explored how ensembling across spatially distributed transmitter receiver links can improve robustness [6], and Akpabio and Bulut applied WiFi sensing to monitoring therapeutic robotic systems in PhysiFi [7], both illustrating the breadth of settings where CSI based sensing is being pushed toward real deployments rather than controlled lab studies.

Two recent works speak directly to the difficulty of sensing from communication traffic rather than a dedicated probing signal. Hu et al. proposed a structured, multilayered framework for evaluating the quality of CSI, assessing packet loss and amplitude consistency before determining whether a capture is suitable for a given sensing task [8]. Dong, Yang, and Srivastava introduced UniFi, an integrated sensing and communication framework that learns directly from the irregularly sampled, heterogeneous CSI produced by ordinary communication packets rather than injecting dedicated probes [9]. Both works confirm that communication generated CSI is fundamentally messier than actively probed CSI, but neither connects this messiness back to the type of traffic producing it. This paper's contribution is precisely that connection, measuring how specific, everyday traffic types, streaming in particular, degrade passive sensing accuracy, and identifying packet starvation as a concrete mechanism behind that degradation.

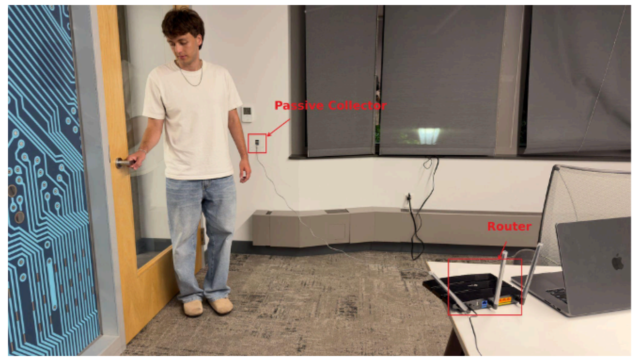

Fig. 1. Data collection setup, showing the router, passive collector, and sensing target used across both active and passive captures.

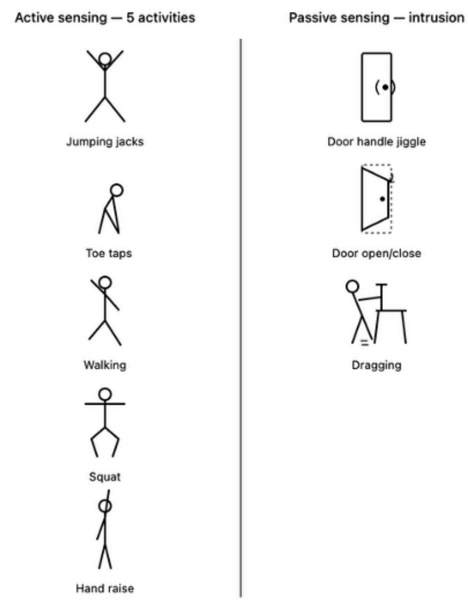

Fig. 2. Activities recognized under active sensing (whole body motion) and under passive, intrusion focused sensing.

III. EXPERIMENTAL SETUP

Both sensing modes were built around an ESP32 microcontroller running the ESP32 CSI Tool. For active sensing, the ESP32 was configured as a transmitter receiver (TX-RX) link with a fixed, evenly spaced probing rate. For passive sensing, a second ESP32 was reflashed into a passive sniffer mode and placed near an ordinary Wi-Fi router, listening to whatever traffic the router happened to be carrying rather than transmitting anything itself. The router was configured through a scripted scenario generator to produce controlled background traffic conditions, including browsing, uploading, downloading, streaming, and idle periods, so that the same traffic mixture could be reproduced across recordings.

The active sensing task covered five whole body activities: jumping jacks, toe taps, walking, squats, and hand raises. The passive sensing task targeted three intrusion relevant behaviors: a door handle being jiggled, a door being opened and closed, and an object being dragged along the floor. Each activity was repeated multiple times in randomized order, with short rest periods between repetitions that also served as per cycle baseline segments for later local baseline subtraction. Both the active and passive recordings were collected in the same room with the same equipment layout, illustrated in Fig. 1 and Fig. 3, which removes location and multipath differences as a confound when comparing the two sensing modes directly. Mixed background traffic was active throughout every passive capture to keep the evaluation representative of a real, in use network rather than an artificially quiet one.

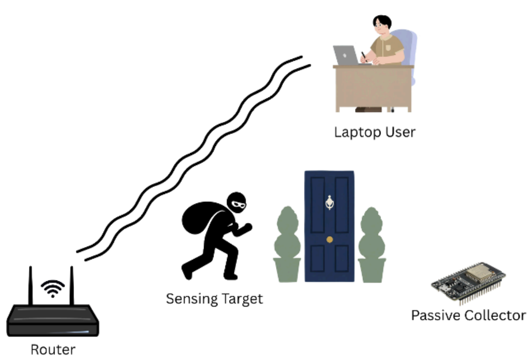

Fig. 3. Sensing environment overview: router, sensing target, laptop user, and passive collector within the shared testbed.

IV. SENSING PIPELINE

The two sensing modes share the first processing steps but diverge once packet timing becomes the dominant source of variability. Both pipelines begin by capturing raw CSI and extracting per subcarrier amplitude, followed by a denoising stage that removes hardware and environmental noise. From there, the controlled active sensing pipeline reduces the denoised amplitude features with principal component analysis (PCA) and classifies them with a deep neural network / convolutional neural network (DNN / CNN), since the active link's steady, evenly spaced packet stream makes a fixed feature representation reliable.

The communication aware passive sensing pipeline instead moves from denoising into a time based windowing stage, where each window is baseline subtracted and z-score normalized on a per session basis before being windowed in time rather than by a fixed packet count. This is then passed to an attention based classifier, which lets the model weight the packets that actually arrive during a window rather than assuming an evenly spaced sequence. This adjustment was necessary because passive packet arrival is dictated entirely by the surrounding network traffic rather than by the sensing system itself, a distinction quantified in Section V.

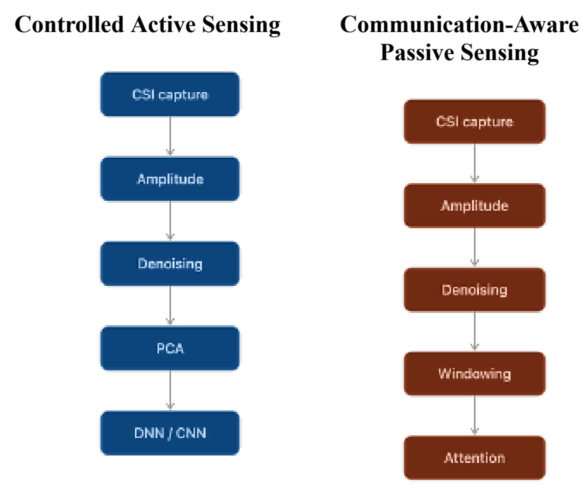

Fig. 4. Active sensing pipeline (left) and communication aware passive sensing pipeline (right); the passive chain replaces PCA with time based windowing and attention to cope with irregular packet arrival.

V. RESULTS

This section reports results for the two classification tasks introduced above, active sensing based general activity recognition and passive sensing based intrusion behavior detection, and then examines how passive sensing accuracy changes under different network traffic conditions.

A. Active Sensing: General Activity Recognition

The active sensing pipeline reached an overall accuracy of 81% across the five whole body activities (macro averaged precision, recall, and F1 all at 0.81, over 9,289 test windows). Table II breaks this down by class. Toe taps and squats were recognized most reliably, with recall of 95% and 89% respectively, since their footwork produces a distinctive and repeatable amplitude signature. Jumping jacks were the weakest class at 68% F1, most often confused with hand raises, which makes sense given that both activities involve similar overhead arm motion and differ mainly in the accompanying leg movement, a distinction that is harder for a single antenna link to resolve. The corresponding confusion matrix is shown in Fig. 5.

TABLE II. Classification Report — General Activity Recognition (Active Sensing)

Class	Precision	Recall	F1-score	Support
Hand_Raise	0.80	0.74	0.77	1958
Jumping_Jacks	0.66	0.71	0.68	1891
Squat	0.89	0.89	0.89	1724
Toe_Tap	0.92	0.95	0.93	1841
Walking	0.80	0.77	0.79	1875
Accuracy			**0.81**	**9289**
Macro avg	**0.81**	**0.81**	**0.81**	**9289**

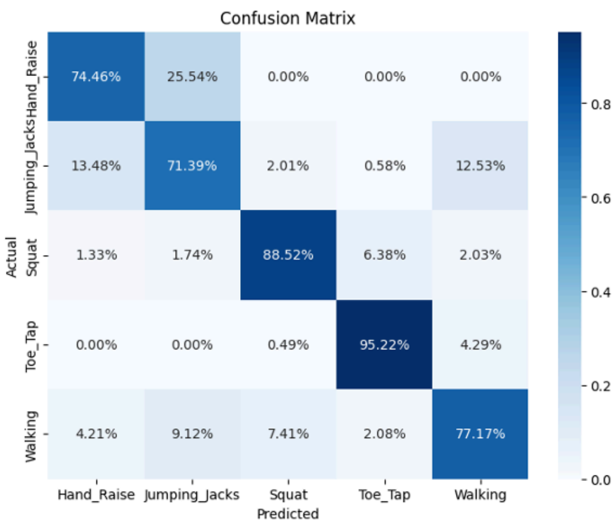

Fig. 5. Confusion matrix for general activity recognition under active sensing (81% accuracy).

B. Passive Sensing: Intrusion Behavior Detection

Using only ambient Wi-Fi packets and no dedicated transmitter near the target, the passive, communication aware pipeline reached an overall accuracy of 75% (macro F1 0.73) across 11,016 test windows, detecting door handle jiggling, door opening and closing, and dragging. Table III shows the per class breakdown. Door opening and closing was detected most reliably, with 89% recall, since it produces a large, sustained change in the channel as the door swings through the signal path. Door handle jiggling and dragging were weaker, with recall between 61% and 72%, and were confused with each other more often than with door opening and closing, since both are subtler, sustained motions that occupy a similar part of the amplitude and timing space. The corresponding confusion matrix is shown in Fig. 6.

TABLE III. Classification Report — Intrusion Behavior Detection (Passive Sensing)

Class	Precision	Recall	F1-score	Support
Door_Handle_Jiggle	0.76	0.61	0.68	—
Door_OpenClose	0.82	0.89	0.85	—
Dragging	0.62	0.72	0.67	—
Accuracy			**0.75**	**11016**
Macro avg	**0.73**	**0.74**	**0.73**	**11016**

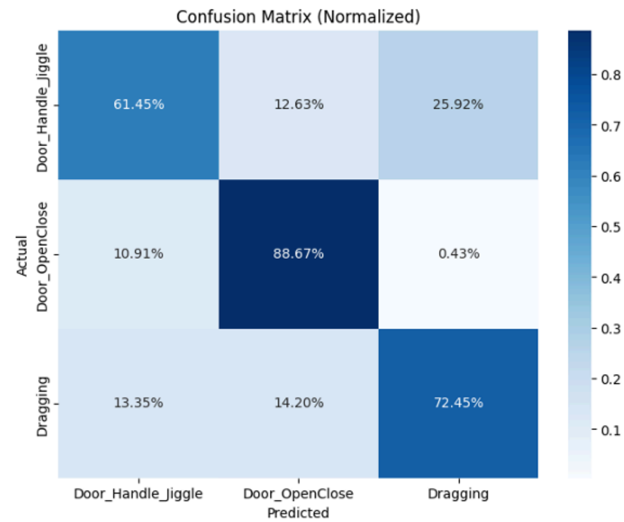

Fig. 6. Confusion matrix for intrusion behavior detection under passive sensing (75% accuracy).

C. Impact of Traffic Conditions on Passive Sensing

The central question of this paper is how much background network traffic actually matters for passive sensing accuracy. Table I summarizes the structural difference between the two sensing modes that makes this question meaningful in the first place. The active TX-RX link held a stable packet rate near 100 packets per second throughout every recording because it controls its own transmission timing. The passive sniffer's packet rate instead ranged from 14 to 241 packets per second across recordings, bursty and entirely dictated by whatever traffic the network happened to be carrying at that moment.

TABLE I. Active vs. Passive Sensing Characteristics

	Active Sensing	Passive Sensing
Packet rate	~100 pps, stable	14–241 pps, bursty
Signal timing	Controlled by TX	Dictated by traffic
Depends on network?	No	Yes

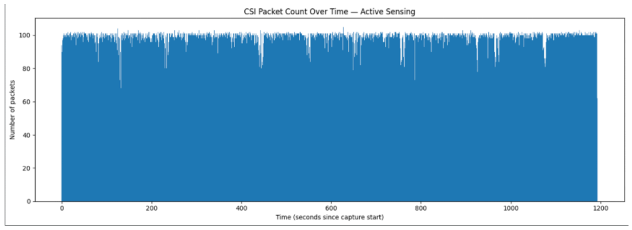

Fig. 7. CSI packet count over time under active sensing, held near a stable ~100 packets per second.

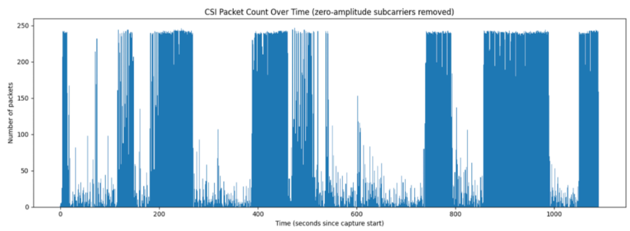

Fig. 8. CSI packet count over time under passive sensing with mixed background traffic; packet arrival tracks network activity rather than the sensing target.

TABLE IV. Leave-One-Recording-Out Accuracy by Traffic Condition (Passive Sensing)

Traffic Condition	Mean Accuracy	Mean F1
Active (TX-RX)	56.1% ± 7.5%	54.6% ± 8.8%
Download	45.8% ± 5.9%	43.8% ± 6.7%
Mixed	25.4% ± 12.9%	21.5% ± 11.4%
Streaming	14.2% ± 5.1%	12.9% ± 4.8%

Table IV reports leave one recording out cross validated accuracy for the general activity recognition task, evaluated separately under each traffic condition using the same passive sensing pipeline throughout. Accuracy fell sharply and consistently as background traffic shifted from download (45.8%) to a mixed traffic profile (25.4%) to streaming (14.2%), well below the active sensing baseline of 56.1% collected in the same room with the same equipment. This ordering is not a modeling artifact. Raw CSI packet counts collected over the same duration and room tell the same story before any processing takes place: the active link captured roughly 642,000 packets, download traffic yielded about 110,000, mixed traffic about 67,000, and streaming traffic only about 63,000, an order of magnitude fewer than active sensing captured in the same window of time.

Streaming traffic is bandwidth heavy and bursty in a way that crowds out the smaller, more frequent packets a passive sniffer relies on to reconstruct a fine grained picture of channel state. The result is a second, distinct mechanism by which traffic degrades passive sensing beyond ordinary signal noise: data starvation, where the sensor simply does not see enough packets to work with, regardless of how good its downstream model is. This distinction matters for anyone deploying a passive sensing system, since it means accuracy cannot be improved by model changes alone when the underlying issue is that too few packets are arriving in the first place.

VI. DISCUSSION

These results carry a direct implication for anyone deploying passive Wi-Fi sensing outside a lab: accuracy cannot be treated as a fixed property of a trained model, because it depends on what else is happening on the network at the moment of sensing. A security system built around passive intrusion detection may perform well during quiet, idle network periods and degrade substantially the moment a resident starts streaming video, precisely when reliable detection matters most. Active sensing sidesteps this problem entirely by controlling its own transmission timing, which is the practical tradeoff between the two paradigms explored here: passive sensing needs no dedicated hardware near the target and is easier to deploy discreetly, while active sensing is far more robust to whatever else the network is doing.

This study was conducted in a single indoor testbed with one router configuration, so the exact accuracy figures reported here should be read as evidence of a real and substantial effect rather than as universal constants that will hold in every environment. A natural next step is a traffic aware CSI quality metric, one that scores an incoming window by how much usable sensing signal likely survived the traffic conditions present at capture time, so that a deployed system can flag low confidence windows rather than silently degrading.

VII. CONCLUSION

This paper presented a systematic, communication aware study of how everyday network traffic affects passive Wi-Fi sensing. By building active and passive sensing pipelines on the same ESP32 based platform and evaluating them in the same physical space, we isolated traffic type as a distinct and substantial source of accuracy degradation in passive sensing, separate from the sensing algorithm itself. Streaming traffic in particular starves the passive receiver of the packets it needs, an effect visible both in raw packet counts and in downstream classification accuracy. Applying the same passive pipeline to a security relevant intrusion detection task showed that reliable behavior recognition from ambient signals alone is achievable, but that its reliability is itself a function of network conditions that deployed systems will need to account for. Future work should extend this evaluation across more environments and traffic mixes, and move toward sensing systems that reason explicitly about the quality of the traffic they are given.

ACKNOWLEDGMENT

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